PROJECT REPORT

AI LEGAL DOCUMENT ANALYZER

Intelligent Contract Analysis and CUAD Semantic Matching

Submitted by

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Abstract

The AI Legal Document Analyzer is a web-based application that processes legal documents and organizes important contract information. It accepts PDF, DOCX and TXT files and applies natural language processing and rule-assisted semantic matching to present summaries, entities, clauses, obligations, potential risks, dates and selected CUAD contract categories. This report includes the supplied screenshots of the interface, sample analysis and deployment.

1. Introduction

Legal agreements can be lengthy and difficult to review manually. This project provides an assistive first-pass analysis by extracting text and organizing selected contract details. It is an informational tool and does not replace a qualified lawyer.

2. Objectives

- Provide a web interface for uploading legal documents.
- Support PDF, DOCX and TXT formats.
- Generate a structured contract summary.
- Extract entities, clauses, obligations and dates.
- Highlight possible omissions through rule-based risk checks.
- Compare document sentences with 41 CUAD contract-category descriptions.

3. Technologies and Method

The project uses Python and Flask for the web application, HTML/CSS templates for the interface, document-reading utilities for supported file formats, and NLP components for extraction and analysis. CUAD matching compares document sentences with descriptions of 41 contract categories using semantic similarity with rule assistance. The displayed similarity values are experimental matching indicators, not legal confidence scores. The matcher has not been fine-tuned or validated for legal decisions.

4. Main Features

- Document upload: Accepts PDF, DOCX and TXT files.
- AI summary: Presents contract type and selected payment, confidentiality, termination, liability and governing-law information.
- Entity extraction: Displays detected money, duration, dates and geographic entities.
- Clause extraction: Identifies selected termination and liability clauses.
- Obligation extraction: Groups statements into termination/notice, general, liability and payment obligations.
- Legal risk detection: Reports selected provisions not detected, with reason, evidence and recommendation.
- Dates and deadlines: Lists detected dates and durations with context.
- CUAD semantic detection: Shows matched category, sentence, similarity and matching method.

5. Demonstration Results

The supplied result screenshots show analysis of 'cuad_test_contract.docx' with a reported word count of 78. The visible output includes:

- Contract summary for a service agreement.
- Extracted values: INR 500000, INR 10000, 30 days, 31 December 2026 and India.

- Termination statements covering 30 days' written notice and termination for convenience.
- Liability statements covering a stated cap, liquidated damages and unlimited supplier liability.
- Risk flags for confidentiality, payment, dispute resolution and liability-cap provisions not detected by the checks.
- Date/deadline output for 31 December 2026 and a 30-day notice period.
- CUAD matches including Expiration Date, Notice Period To Terminate Renewal, Liquidated Damages, Termination For Convenience and Uncapped Liability.

6. Application Screenshots

The following screenshots document the application's interface and sample analysis output.

Figure 1. Application home page and document upload interface.

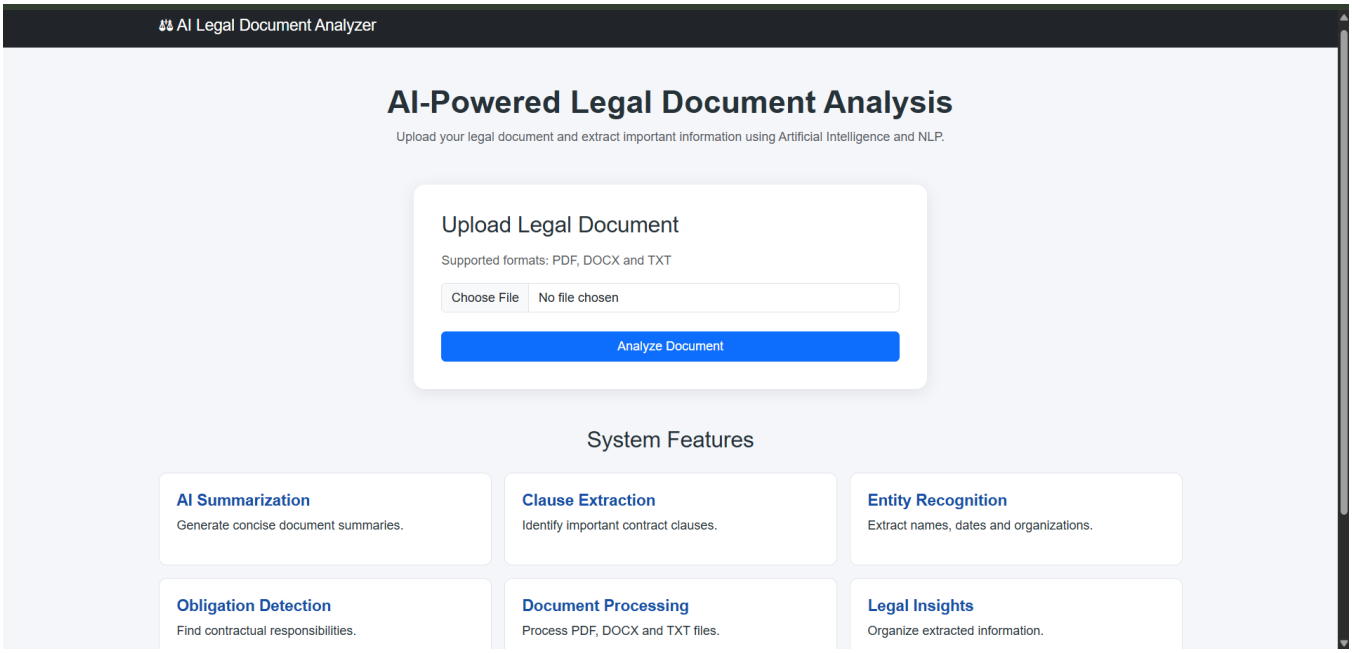


Figure 2. Document information and AI-generated contract summary.

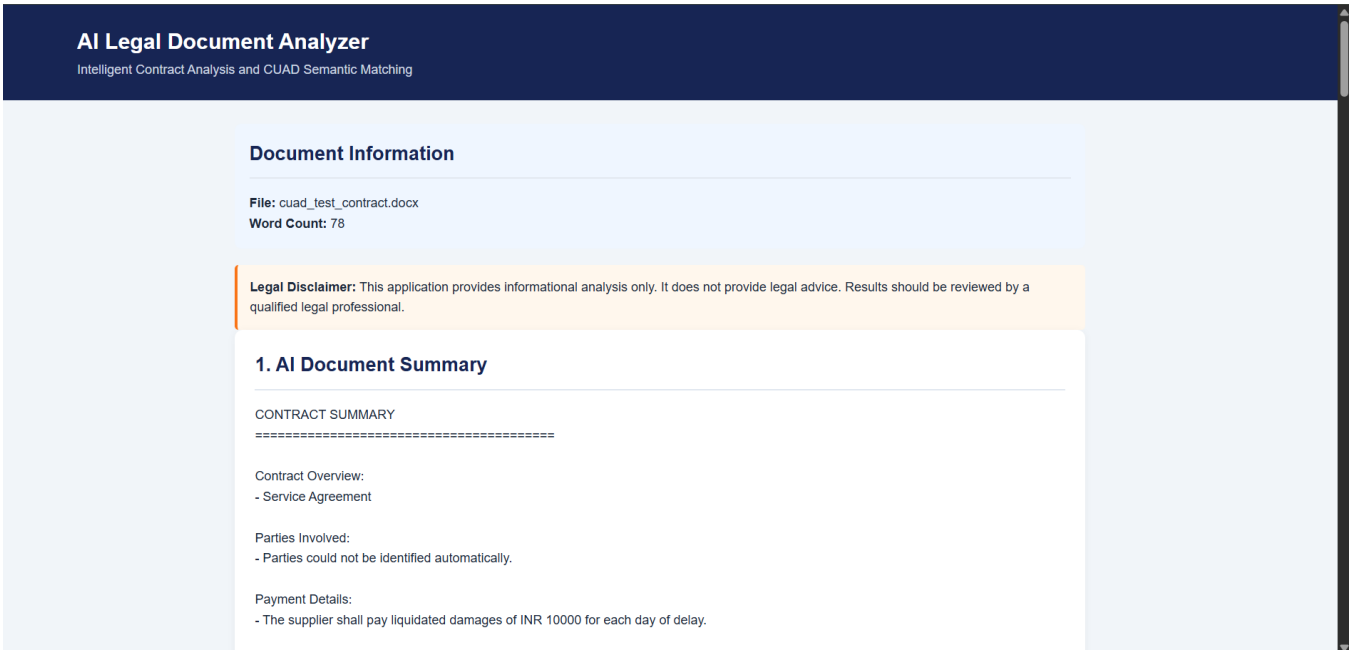


Figure 3. Continuation of the summary, including termination and liability provisions.

Confidentiality:
- Not identified in the document.

Termination Conditions:
- Either party may terminate this agreement by providing 30 days written notice.
- Either party may terminate this agreement for convenience without cause by providing written notice.

Liability Provisions:
- The total liability of either party shall not exceed INR 500000.
- The supplier shall pay liquidated damages of INR 10000 for each day of delay.
- The liability of the supplier shall be unlimited.

Governing Law:
- This agreement shall be governed by the laws of India.

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Note: Automated summary for informational purposes only. Verify all details against the original contract.

Figure 4. Entity extraction results.

2. Entity Extraction

Entity Type	Extracted Value
MONEY	INR 500000
MONEY	INR 10000
DURATION	30 days
DATE	31 December 2026
GPE	India
DATE	each day

Figure 5. Clause extraction results.

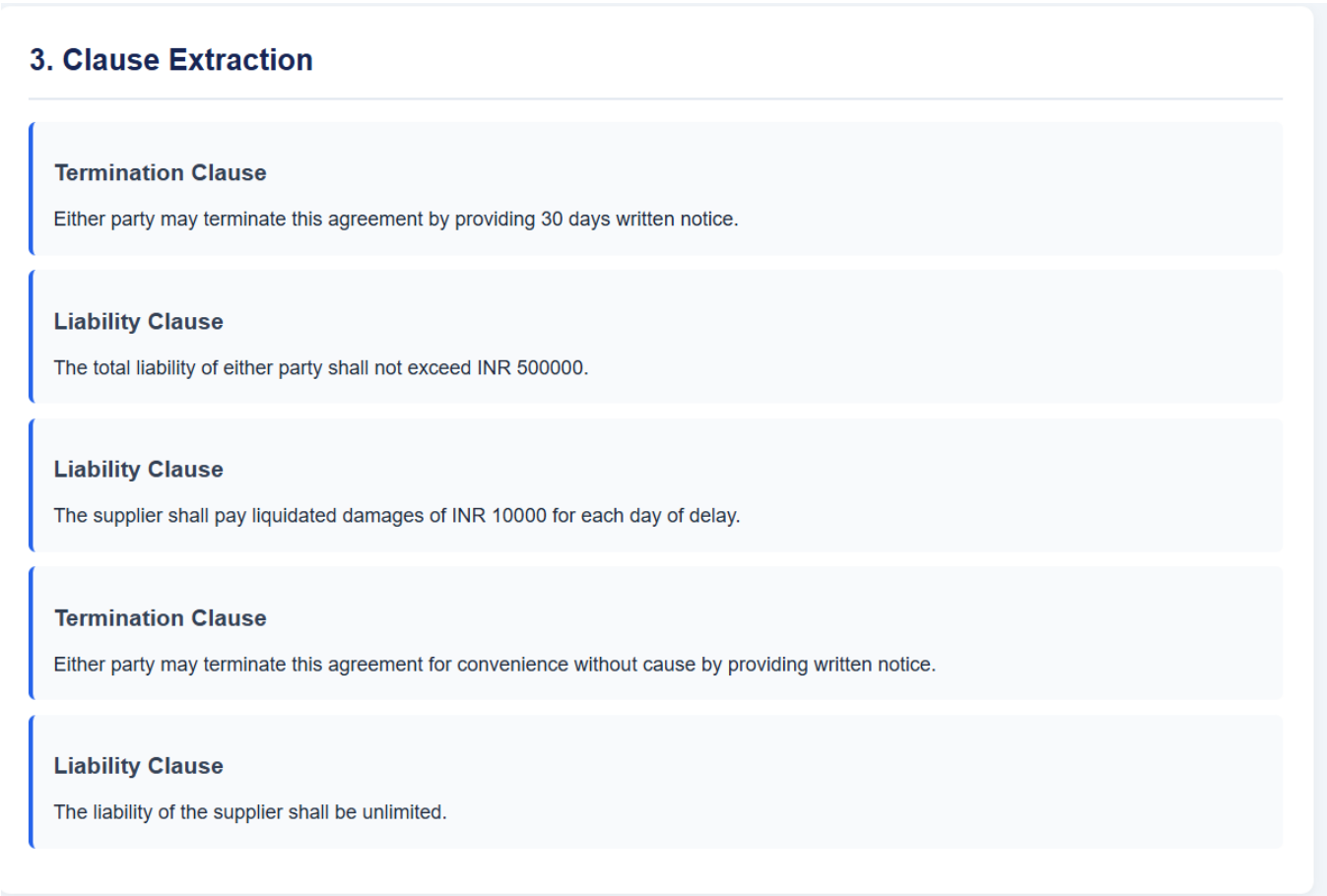


Figure 6. Obligation extraction results.

4. Obligation Extraction

Termination / Notice Rights

Either party may terminate this agreement by providing 30 days written notice.

General Obligation

This agreement shall be governed by the laws of India.

Liability Obligation

The total liability of either party shall not exceed INR 500000.

Payment Obligation

The supplier shall pay liquidated damages of INR 10000 for each day of delay.

Termination / Notice Rights

Either party may terminate this agreement for convenience without cause by providing written notice.

Liability Obligation

The liability of the supplier shall be unlimited.

Figure 7. Legal risk detection results.

5. Legal Risk Detection

Confidentiality Provision Not Detected

Risk Level: MEDIUM
Reason: The document may not explicitly address protection of confidential information.
Evidence: No confidentiality-related wording was identified.
Recommendation: Consider whether confidentiality obligations are needed.

Payment Provision Not Detected

Risk Level: MEDIUM
Reason: Financial obligations may not be clearly documented.
Evidence: No payment-related wording was identified.
Recommendation: Review whether payment amounts, due dates and payment conditions should be included.

Dispute Resolution Not Detected

Risk Level: MEDIUM
Reason: The contract may not clearly explain how disagreements are handled.
Evidence: No dispute resolution wording was identified.
Recommendation: Consider reviewing appropriate dispute resolution procedures.

Liability Cap Not Detected

Risk Level: MEDIUM
Reason: The agreement may not clearly establish a maximum financial exposure.
Evidence: Liability-related wording exists, but an explicit financial cap was not identified.
Recommendation: Review whether liability limits and appropriate exceptions are required.

Figure 8. Dates and deadlines and the beginning of CUAD semantic detection.

6. Dates and Deadlines

Type	Value	ISO Date	Context
Date	31 December 2026	2026-12-31	This agreement expires on 31 December 2026.
Duration	30 days	Not applicable	Either party may terminate this agreement by providing 30 days written notice.

7. CUAD Semantic Clause Detection

This module compares document sentences with descriptions of 41 CUAD contract categories.

Expiration Date

Matched Sentence

SERVICE AGREEMENT This agreement expires on 31 December 2026.

Similarity: 0.698 | Matching Method: rule-assisted semantic

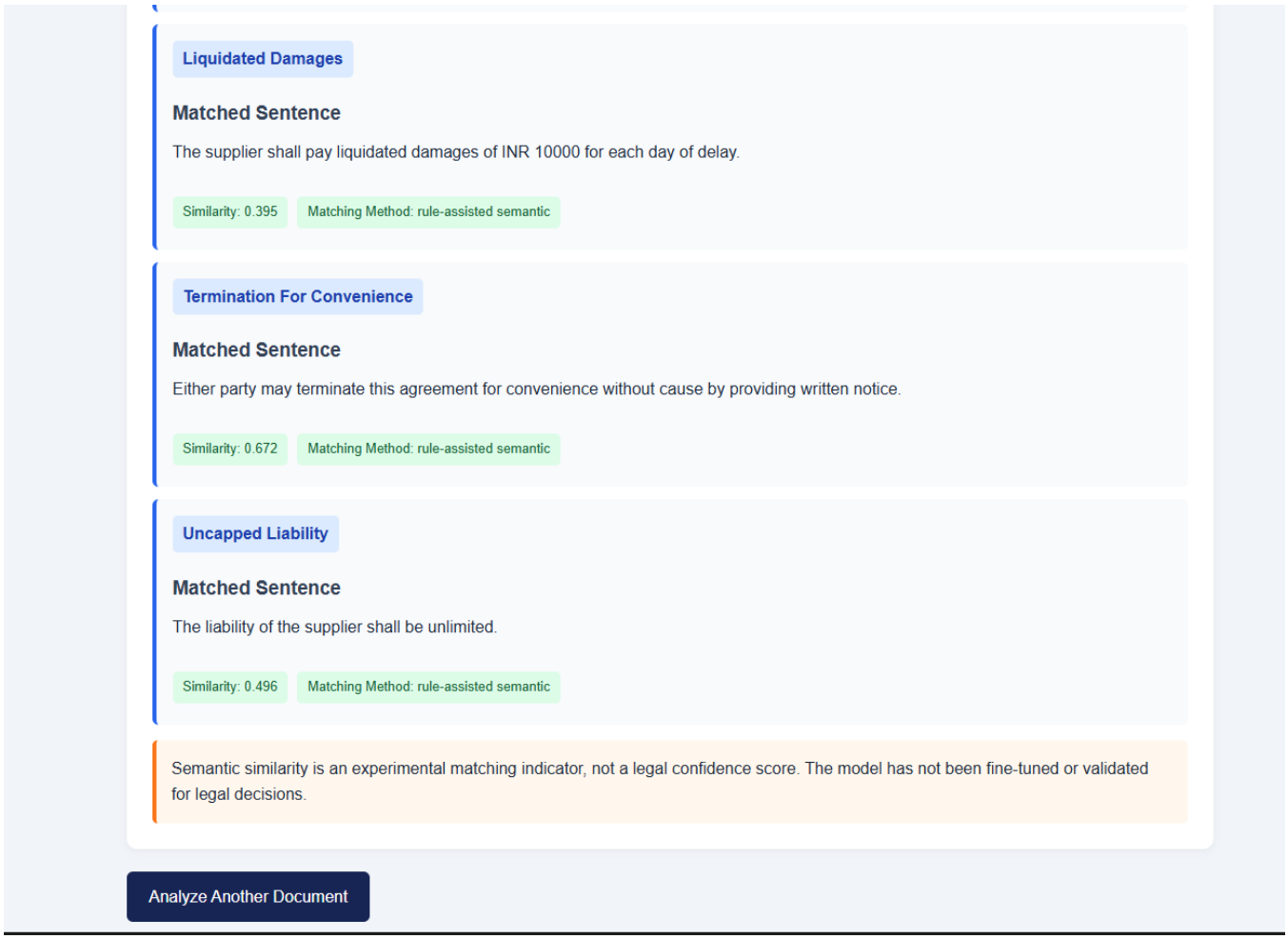
Notice Period To Terminate Renewal

Matched Sentence

Either party may terminate this agreement by providing 30 days written notice.

Similarity: 0.542 | Matching Method: rule-assisted semantic

Figure 9. Further CUAD matches and the experimental matching disclaimer.



7. Deployment Status

The supplied Render log shows Gunicorn starting and listening on the configured port, followed by successful health-check requests. A browser screenshot also shows the application loading. However, a later screenshot displays HTTP 502 Bad Gateway. The evidence therefore shows that the service responded successfully at one point but subsequently experienced an availability problem.

A 502 response generally means the hosting gateway could not obtain a valid response from the application process. The visible log excerpt does not establish the root cause. Complete Render logs around the failure are needed to identify a traceback, worker restart, memory termination or timeout.

8. Limitations

- Automated extraction may miss entities or misinterpret context.
- A provision marked 'not detected' is not proof that the contract lacks it.
- Semantic similarity is experimental and is not a legal confidence score.
- The screenshots demonstrate one sample document and do not establish accuracy across all contracts.
- Deployment availability requires further investigation because an HTTP 502 was observed.

9. Future Enhancements

- Evaluate extraction modules using labelled test documents.
- Add export of analysis results to PDF or DOCX.

- Improve upload validation and secure document handling.
- Improve deployment logging, monitoring and reliability.
- Evaluate semantic matching on held-out legal documents before high-stakes use.

10. Conclusion

The AI Legal Document Analyzer demonstrates a workflow for uploading legal documents and organizing selected contract information through summarization, extraction, risk checks and CUAD semantic matching. The screenshots document the interface and sample output. The observed 502 response indicates that deployment reliability still needs investigation. All findings should be checked against the original contract and reviewed by a qualified legal professional.

Legal Disclaimer

This application provides informational analysis only and does not provide legal advice. Automated results may be incomplete or inaccurate. Verify all details against the original document and consult a qualified legal professional before making legal or commercial decisions.

Programme: B.Tech – Computer Science and Engineering

Academic Year: Third Year

Institution: SRM University-AP

Project: AI Legal Document Analyzer

GitHub Repository: https://github.com/AP24110011282/AI-Legal-Document-Analyzer

Deployment URL: https://ai-legal-document-analyzer-c1jq.onrender.com